



COM SATS University Islamabad, Lahore Campus

Department of Mathematics

COURSE HANDBOOK

1	Course Title	Linear Algebra
2	Course Code	MTH 231
3	Credit Hours	3(3,0)
4	Semester	SP2024
5	Resource Person	Dr. Imran Ahmed
6	Supporting Team Members	Will be informed later
7	Contact Hours (Theory)	3 hours per week
8	Contact Hours (Lab)	Not Applicable
9	Office Hours	Shall be communicated later
10	Course Introduction	
This course is an introduction to Linear Algebra and to some of its significant applications. It is designed for a course at the freshman level. This course introduces many applications of linear algebra in other areas of mathematics, physics, economics, engineering, and computer sciences. Linear Algebra is the undergraduate course that will have the more impact on student's computing skills through conceptual learning. The emphasis is on the computational and geometrical aspects of the subject, keeping abstraction to a minimum.		
11	Learning Objective	
The main objective of this course is to help the students to learn the basic ideas of linear algebra and to see some of its applications. By the end of this course, students should be able to: • Understand some applications of systems of linear equations. • Perform the operations of addition, scalar multiplication, and multiplication, and find the transpose and inverse of a matrix. • Calculate determinants using row operations, column operations, and expansion down any column and across any row. • Prove elementary statements concerning the theory of matrices and determinants. • Understand about vector addition, scalar multiplication, inner products, projections, norms, orthogonal vectors, linear independence, spanning sets, subspaces, bases, and dimension for R_n and abstract vector spaces. • Write the relationships between A being invertible, $\det A$, $AX = 0$ having a solution, the rank of A, and the rows of A being linearly independent. • Use the Gram-Schmidt process to orthogonalize matrices. • Find the kernel, range, rank, and nullity of a linear transformation. • Find the matrix associated with a linear transformation with respect to given bases, and understand the relationship between the operations on linear transformations and their corresponding matrices. • Find the change-of-basis matrix. • Understand the concept of linear transformations. • Calculate eigenvalues and their corresponding eigenspaces. • Determine if a matrix is diagonalizable, and if it is, diagonalize it.		
12	Course Contents	
System of Linear Equations and matrices, determinants; vector and inner product spaces, matrix representations of Linear transformations, eigenvalues and eigenvectors		

Diagonalization of symmetric matrices, Cayley-Hamilton Theorem.

13	Lecture/Lab Schedule
Lecture	Topics
Week 1	Matrices and system of linear equations - System of Linear equations ($Ax = b$) - Homogenous system of Linear equations ($Ax = 0$) - Matrix, - Matrix Operations and row equivalent matrices - Algebraic properties of matrix operations
Week 2	Properties of matrices with its applications - Partitioned matrices - Special properties of matrix, - Echelon form - Reduced Echelon form
Week 3	Methods of solving systems of linear equations - Gauss-Jordan elimination method - Gauss-Jordan reduction method
Week 4	Determinant and methods of finding inverse of a matrix - Singular matrices and non-singular matrices - Method for finding inverse via row operations. - Introduction to determinants, - Properties of determinants
Week 5	Adjoint of a matrix and Cramer's rule - Cofactor expansion and adjoint of a matrix - The inverse of a matrix using determinants - Cramer's Rule
Week 6	Vector spaces - Introduction to binary operations - Introduction to vector spaces - Subspaces.
Week 7	Introduction to basis of a vector space - Linear Dependence and Independence. - Linear spanning - Basis and Dimensions - Computing a spanning set
Week 8	Basis and Dimensions
Week 9	Coordinate vectors - Coordinate vectors - Matrix of coordinate change of basis.
Week 10	Vector spaces associated to a matrix and rank of a matrix - Row space - Column space - Rank of a matrix - Dimension of a row/column space - Null space and Nullity
Week 11	Inner product and its applications - Introduction to inner product - Properties of inner product - Orthogonal and orthonormal sets - The Gram-Schmidt process.
Week 12	Eigenvalues and Eigenvectors - Eigenvalues and Eigenvectors - Eigen space and basis of eigen space - Similar Matrices

	iv. Cayley Hamilton theorem
Week 13	Diagonalization and its applications
Week 14	Linear transformation and their properties i. Linear transformation from $R^n \rightarrow R^m$. ii. Properties of linear transformations iii. Matrix of linear transformations
Week 15	Kernel and Range of Linear Transformations i. Kernel of linear transformation ii. Range of linear transformation iii. Dimension theorem

14 Course Assessment

The assessment of this module shall have following breakdown structure

Midterm Examination	25%
Quizzes/Assignments	25%
Terminal Examination	50%

The minimum pass marks for each course shall be 50%. Students obtaining less than 50% marks in any course shall be deemed to have failed in that course. The correspondence between letter grades, credit points, and percentage marks at CIIT shall be as follows:

Grades	Letter Grade	Credit Points	Percentage Marks
A	(Excellent)	4.00	85 and above
A-		3.66	80-84
B+		3.33	75-79
B	(Good)	3.00	71-74
B-		2.66	68-70
C+		2.33	64-67
C	(Average)	2.00	61-63
C-		1.66	58-60
D+		1.30	54-57
D	(Minimum passing)	1.00	50-53
F	(Failing)	0.00	Less than 50

15 Assessment Schedule

3 rd week of September	1 st Assignment and Quiz
3 rd week of October	2 nd Assignment and Quiz
3 rd week of November	3 rd Assignment and Quiz
2 nd week of December	4 th Assignment and Quiz

16. Format of Assignment

This course indoctrinates the following format for all its assignments:

- Paper Size: A4
- Left Margin: 2 Inches
- Right Margin: 1 Inch
- Top Margin: 0.5 Inch
- Bottom Margin: 0.5 Inch
- Font: Times New Roman
- Font Size:
 - Main Heading 14
 - Sub Heading 12
 - Text 12
 - Titles 16
- Font Color: Black
- Line Spacing: 1.5

10. Diagrams & Charts: Need not be colored 11. Title page must be designed as guided by resource person in class 12. Number of Pages: No Limit 13. Reference Style: APA (If applicable)		
17.	Text Book	Introduction to Linear Algebra, by B. Kolman (7 th Edition)
18.	Reference Books	1. Elementary Linear Algebra, by Anton (8 th Edition) 2. Linear Algebra and its applications, by David C Lay (3 rd Edition)
19.	Plagiarism Plagiarism involves the unacknowledged use of someone else's work, usually in coursework, and passing it off as if it were one's own. Many students who submit apparently plagiarised work probably do so inadvertently without realising it because of poorly developed study skills, including note taking, referencing and citations; this is poor academic practice rather than malpractice. Some students, particularly those from different cultures and educational systems, find UK academic referencing/acknowledgement systems and conventions awkward, and proof-reading is not always easy for dyslexic students and some visually-impaired students. Study skills education within programmes of study should minimise the number of students submitting poorly referenced work. However, some students plagiarise deliberately, with the intent to deceive. This intentional malpractice is a conscious, pre-mediated form of cheating and is regarded as a particularly serious breach of the core values of academic integrity. Plagiarism can include the following: - collusion, where a piece of work prepared by a group is represented as if it were the student's own; - commission or use of work by the student which is not his/her own and representing it as if it were, e.g.: - purchase of a paper from a commercial service, including internet sites, whether pre-written or specially prepared for the student concerned - submission of a paper written by another person, either by a fellow student or a person who is not a member of the university; - duplication (of one's own work) of the same or almost identical work for more than one module; - the act of copying or paraphrasing a paper from a source text, whether in manuscript, printed or electronic form, without appropriate acknowledgement (this includes quoting directly from another source with a reference but without quotation marks); - submission of another student's work, whether with or without that student's knowledge or consent; - Directly quoting from model solutions/answers made available in previous years; - cheating in class tests, e.g. - when a candidate communicates, or attempts to communicate, with a fellow candidate or individual who is neither an invigilator or member of staff - copies, or attempts to copy from a fellow candidate - attempts to introduce or consult during the examination any unauthorised printed or written material, or electronic calculating, information storage device, mobile phones or other communication device - personates or allows himself or herself to be impersonated. - Fabrication of results occurs when a student claims to have carried out tests, experiments or observations that have not taken place or presents results not supported by the evidence with the object of obtaining an unfair advantage. These definitions apply to work in whatever format it is presented, including written work, online submissions, group work and oral presentations.	
20.	Attendance Policy Every student must attend 80% of the lectures/seminars delivered in this course.	
21.	Field Trips/Case Studies/Seminars/Workshop Not Applicable	